EE 354/CE 361/MATH 310 - Introduction to Probability and Statistics

Spring 2026

EE 354/CE 361/MATH 310 L4 section (Spring 2026)

Introduction to Probability and Statistics -

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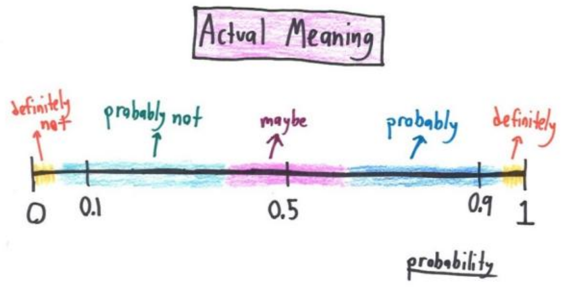
Week 15 - Lecture No 28 – April 20, 2026

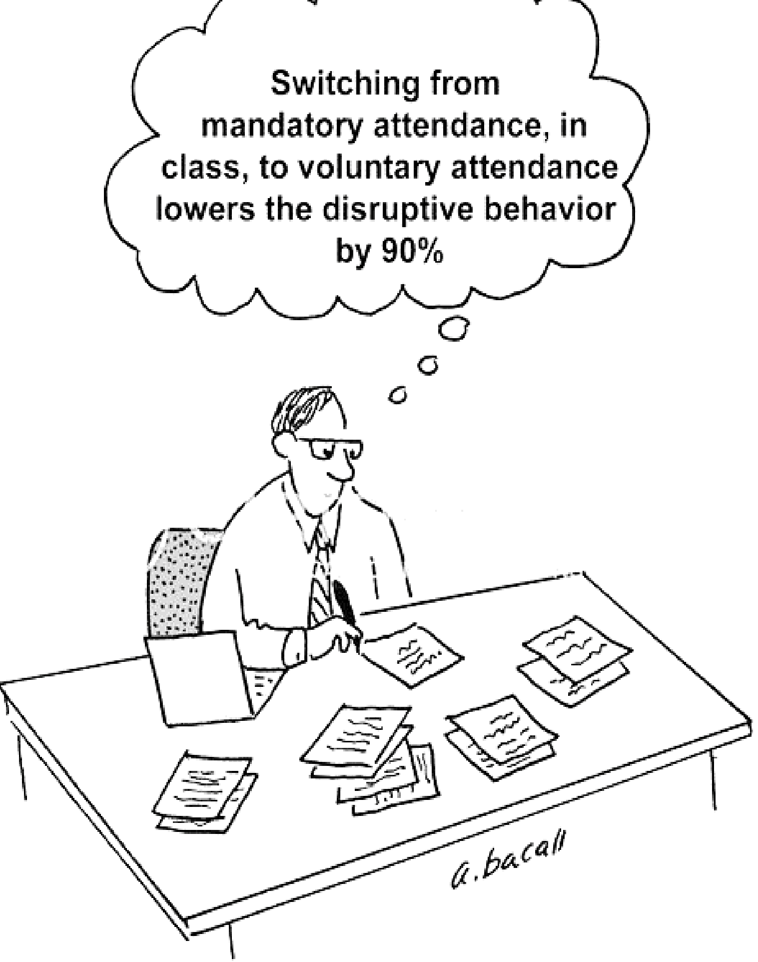


Announcements!

1. No office hours tomorrow - April 21, 2026
2. Quiz No 08 – (pdfs, labels, working)
3. End-term Exam - Monday, May 4, 2026 (8:30a to 10:30a) @ Tariq Rafi, E-220
    1. Cheat Sheet – double-sided A-4 (HAND WRITTEN ONLY notes)
    2. Fall 2025 end-term uploaded
    3. 5 questions – 3 Continuous r.v.s, 1 discrete r.v.s and 1 from chapter 1
3. Planned quizzes
    - Quiz No 09 - April 22, 2026 (Wednesday) – Conditional PDFs
    - Quiz No 10 - April 29, 2026 (Wednesday) – Bayes Rule

Agenda for today

1. Unit 5 – Continuous Random Variable
    - ❑ Conditional PDFs



For the remaining classes you can mark your attendance (anytime) & leave at any time without any intimation

But if you chose to stay then please be with the class

Slide from lecture 1 – Sp 2026
(Fall 2025 Offering)

Course & Instructor evaluations – please!

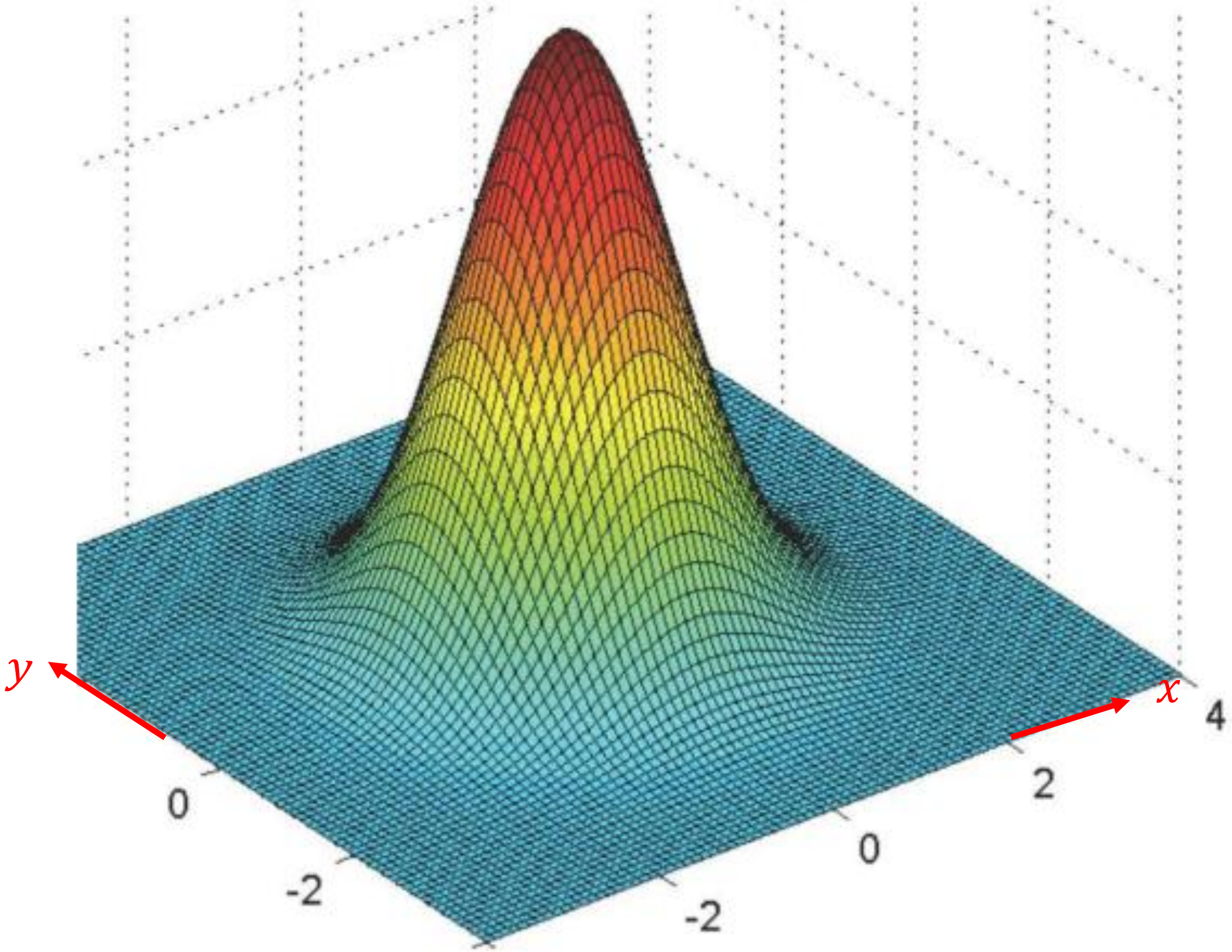
Recap - Conditional PDFs, given another r.v.

$$\mathbf{P}(X \in B \mid Y = y) = \int_B f_{X|Y}(x|y)\,dx$$

- Some comments:

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

  - $f_{X|Y}(x \mid y) \geq 0$



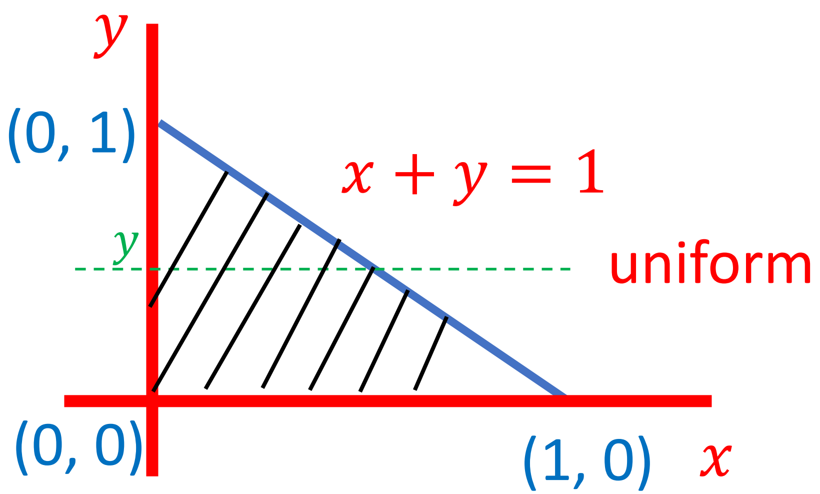
Example 1

Uniform probabilities on a triangle

find:

a) $f_{X,Y}(x, y)$

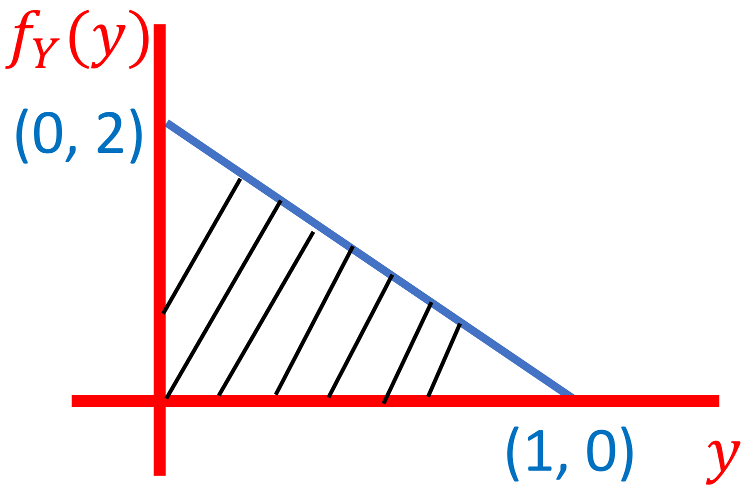
$$f_{X,Y}(x,y) = \begin{cases} 2, & x+y<1,\ x,y>0 \\ 0, & \text{otherwise} \end{cases}$$



b) $f_Y(y)$

$$f_Y(y) = \int f_{X,Y}(x,y)\,dx$$

$$= \begin{cases} 2(1-y), & 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$



Example 1

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

- $f_{X|Y}(x \mid y) \geq 0$

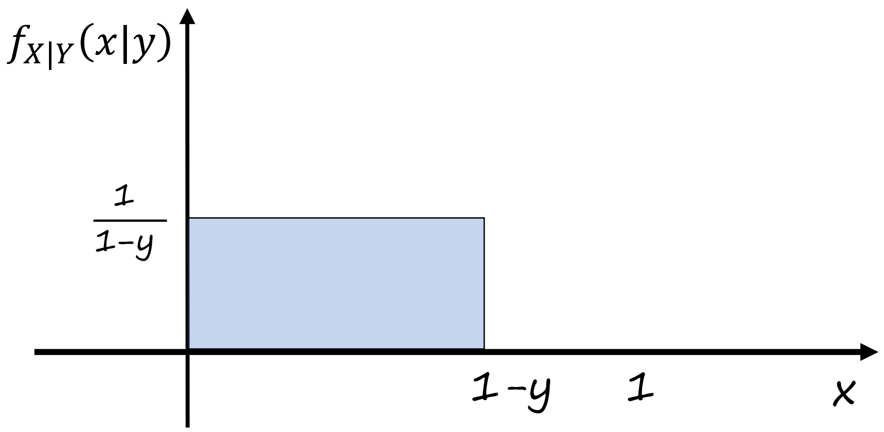
c) $f_{X|Y}(x|y)$

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

$$= \frac{2}{2(1-y)}$$

$$= \begin{cases} \dfrac{1}{(1-y)}, & 0 \leq y \leq 1;\ 0 \leq x \leq 1-y \\ 0, & \text{otherwise} \end{cases}$$

$0 \leq y < 1;$
$0 \leq x + y \leq 1$



Example 1

d) $E[X|Y=y]$

$$E[X \mid Y=y] = \int_{-\infty}^{\infty} x f_{X|Y}(x|y)\,dx$$

$$E[X \mid Y=y] = \int_{0}^{1-y} x \frac{1}{(1-y)}\,dx$$

$$= \frac{1}{(1-y)} \frac{(1-y)^2}{2}$$

$$= \frac{(1-y)}{2}$$

$0 \le y \le 1$

Example 1

e) $E[X]$

$$E[X] = \int_{-\infty}^{\infty} f_Y(y)\, E[X \mid Y=y]\, dy$$

$$= \int_{-\infty}^{\infty} f_Y(y) \frac{(1-y)}{2} dy = \frac{1}{2}\int_{-\infty}^{\infty} f_Y(y) dy - \frac{1}{2}\int_{-\infty}^{\infty} y\, f_Y(y) dy$$

$$= \frac{1}{2} - \frac{1}{2} E[Y] = \frac{1}{2} - \frac{1}{2} E[X]$$

$$\left(1 + \frac{1}{2}\right) E[X] = \frac{1}{2}$$

$$E[X] = \frac{1}{3} \qquad E[Y] = \frac{1}{3}$$

Example 2 - Conditional PDFs (go through this in your own time)

- You have two options to play a game of lottery:

  - Option 1: Select $X \sim U(0, 100)$ then select $Y \sim U(0, X)$. You win Y rupees. Find $f_Y(y)$ and $E[Y]$

  - Option 2: Select $X \sim U(0, 50)$ then select $Y \sim U(0, 2X)$. You win Y rupees. Find $f_Y(y)$ and $E[Y]$

Which option should you select?

Think → Write → Answer

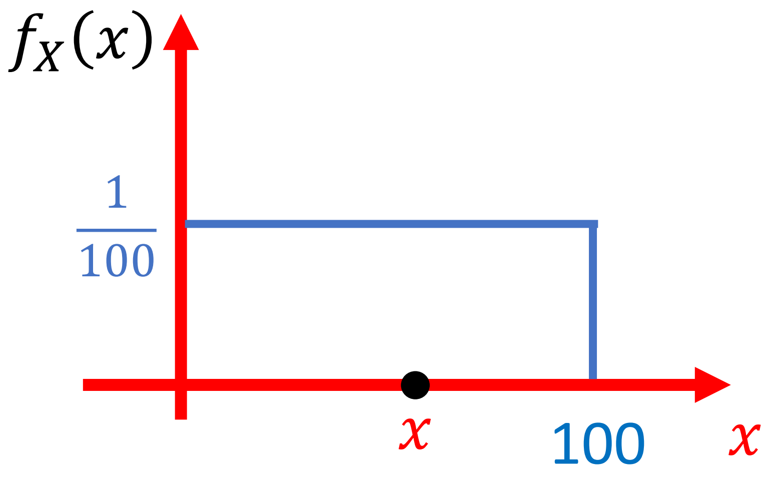
Example 2 - Conditional PDFs

Lets analyze option 1:

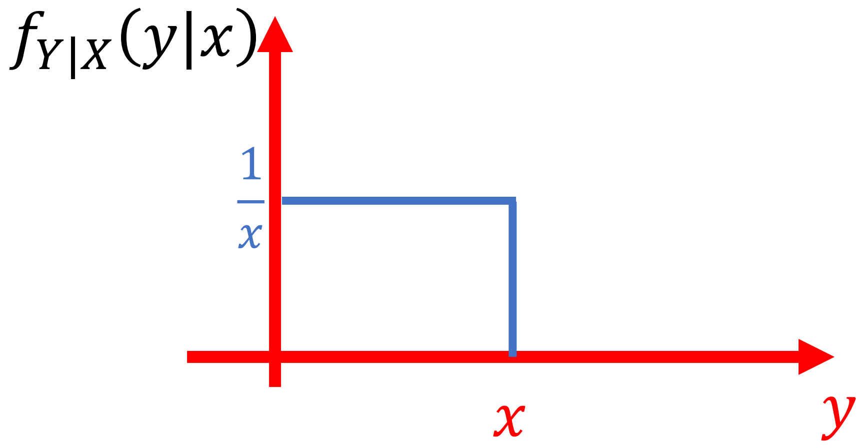
Select x, where $X \sim \mathbf{U}\,(\,\mathbf{0}\,,\mathbf{100}\,)$ then select y such that $Y \sim \mathbf{U}\,(\,\mathbf{0}\,,\mathbf{X}\,)$**.**
You win y rupees. Find $f_Y(y)$ and $E[Y]$

Option 1: Select x, where X ~ U (0 , 100) then select y such that Y ~ U (0 , X).
You win y rupees. Find $f_Y(y)$ and $E[Y]$

Are X & Y independent ? If x = 0.5, this implies y ≤ 0.5

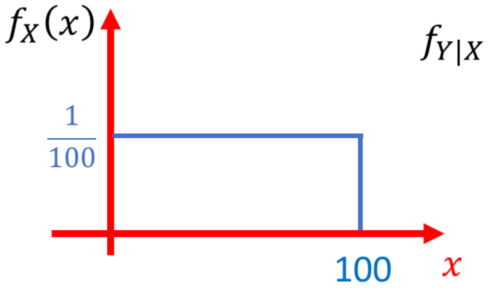


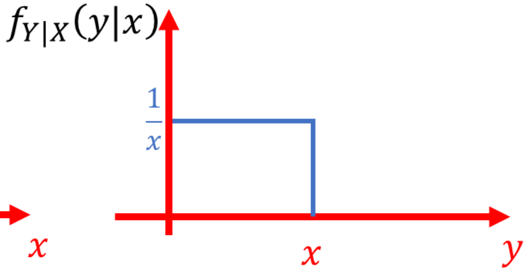
$$f_{X,Y}(x, y) = f_{Y|X}(y|x) f_X(x)$$



Option 1: Select $X \sim U(0, 100)$ then select $Y \sim U(0, X)$. You win Y rupees.

Find $f_Y(y)$ and $E[Y]$





$$f_{X,Y}(x,y) = f_{Y|X}(y|x) f_X(x)$$

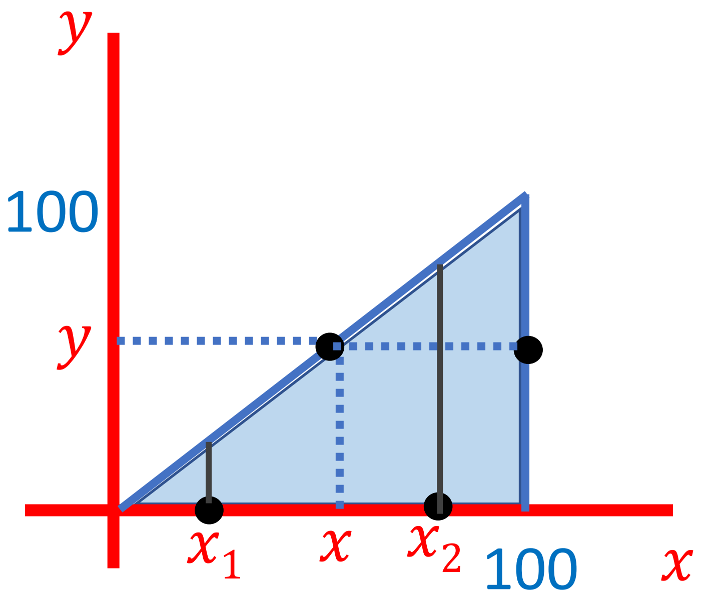
$$= \frac{1}{100} * \frac{1}{x} \qquad 0 \le y \le x \le 100$$

$$f_Y(y) = \int_x f_{X,Y}(x,y)dx = \int \frac{1}{100} * \frac{1}{x} dx$$

$$= \frac{\ln x}{100}\Big|_y^{100} = \frac{1}{100}\ln(100/y) \ ?$$

$$E[Y] = \int_0^{100} y \frac{1}{100}\ln(100/y)\, dy$$

$$= \int_0^{100} E[Y|X=x] f_X(x)dx = \int_0^{100} \frac{1}{100} * \frac{x}{2} dx = 1/2\, E[X] = 25$$



Option 2: Select $X \sim U(0, 50)$ then select $Y \sim U(0, 2X)$. You win Y rupees.

Find $f_Y(y)$ and $E[Y]$

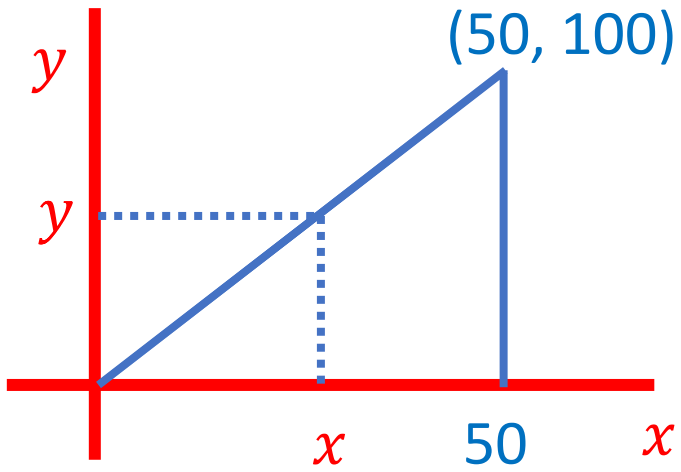
$$f_{X,Y}(x,y) = f_{Y|X}(y|x) f_X(x)$$

$$= \frac{1}{50} * \frac{1}{2x} = \frac{1}{100x}$$

$0 \le x \le 50;\ 0 \le y \le 2x$

$\Downarrow$

$0 \le y \le 100$



$$f_Y(y) = \int_x f_{X,Y}(x,y)dx = \int_{y/2}^{50} f_{X,Y}(x,y)dx$$

$$= \frac{\ln x}{100}\Big|_{y/2}^{50} = \frac{1}{100}\ln(100/y)$$

$$E[Y] = \int_0^{50} y \frac{1}{100} \ln(100/y)\, dy$$

$$= \int_0^{50} E[Y|X=x] f_X(x)dx = \int_0^{50} \frac{1}{50} x\, dx = 1/2\ E[X] = 25$$

Independence

If X and Y are independent

- $\mathbf{E}[\,X\,Y\,] = \mathbf{E}[\,X\,]\,\mathbf{E}[\,Y\,]$
- $\boldsymbol{var}(\mathrm{X}+\mathrm{Y}) = \boldsymbol{var}(\mathrm{X}) + \boldsymbol{var}(\mathrm{Y})$

if g(x) and h(y) are also Indepedent

$\Rightarrow \quad E[\,g(X)\,h(Y)\,] = E[\,g(X)\,]\,E[\,h(Y)\,]$

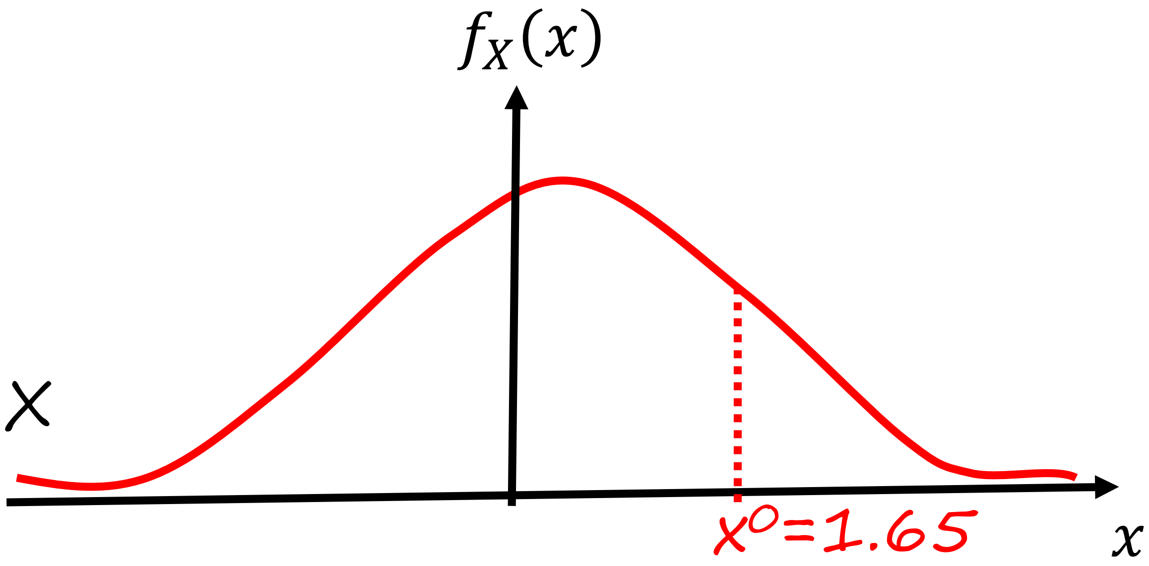
Example:

- Independence in Normal RV's X and Y:

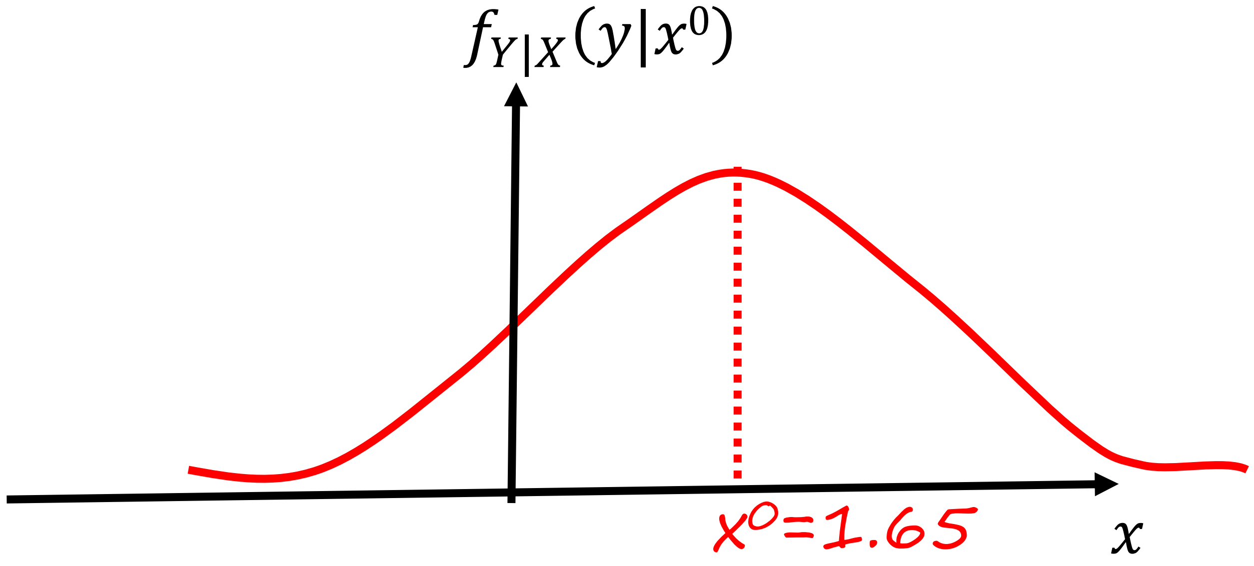
  Lets form a RV in a two step process

1. $X \sim N(0, 1)$;

pick an X



2. $Y \sim N(X, 1)$



are X and Y independent?

Recap - Independent Standard Normal

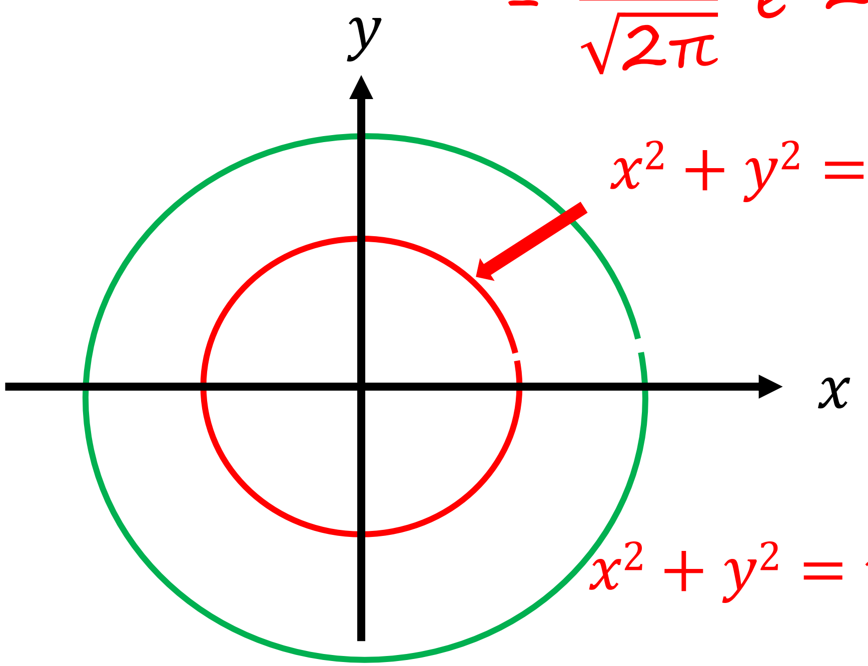
Now let's assume X and Y independent on $\sim N(0, 1)$

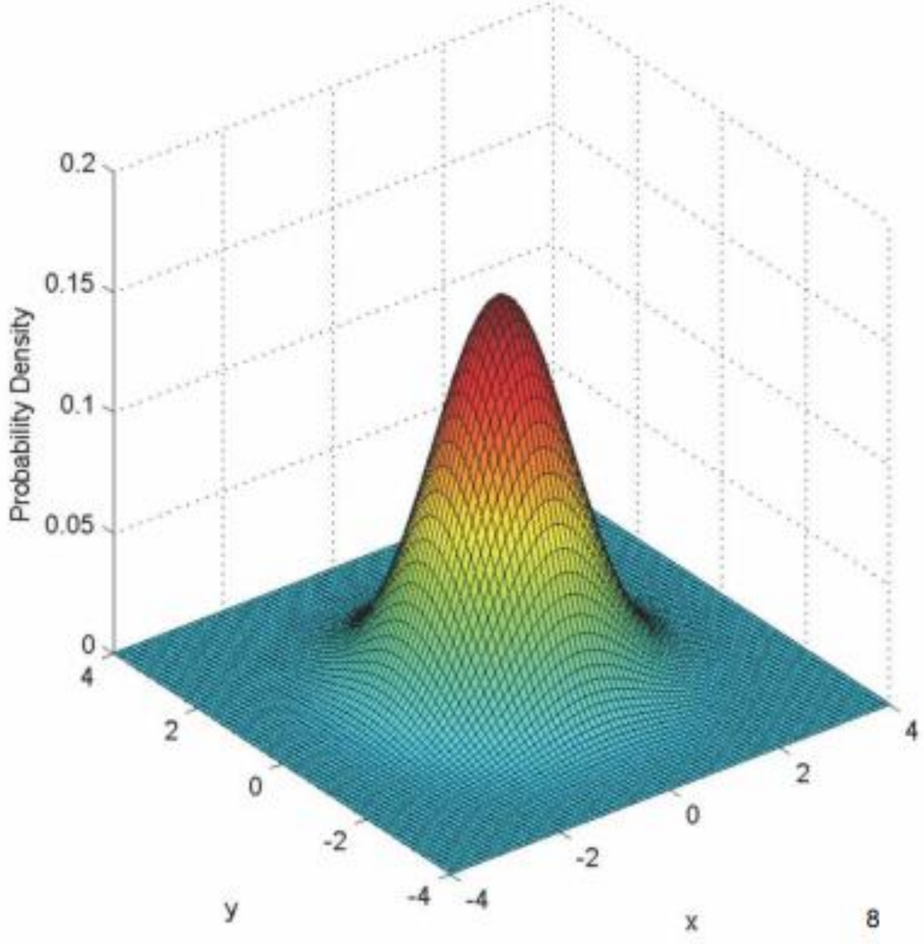
$$f_{X,Y}(x,y) = f_X(x)\ f_Y(y)$$

$$= \frac{1}{\sqrt{2\pi}}\ e^{-\frac{x^2}{2}}\ \frac{1}{\sqrt{2\pi}}\ e^{-\frac{y^2}{2}}$$

$$= \frac{1}{\sqrt{2\pi}}\ e^{-\frac{1}{2}(x^2+y^2)}$$

$\mu_x = \mu_x = 0;\ \ \sigma^2{}_x = \sigma^2{}_y = 1;$



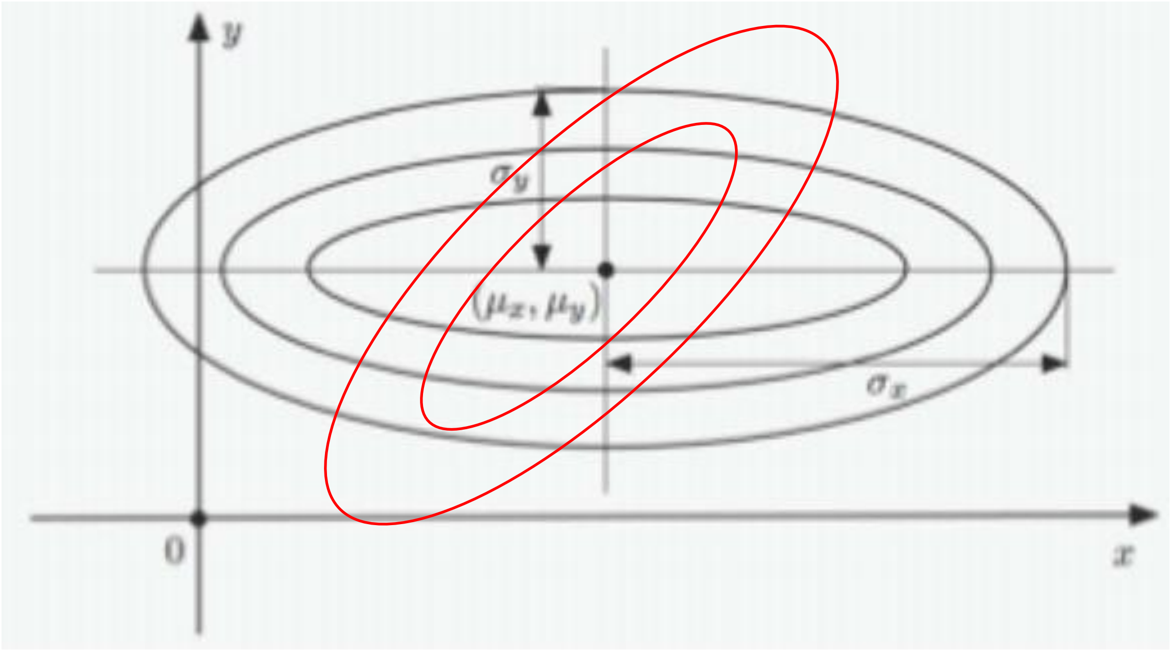


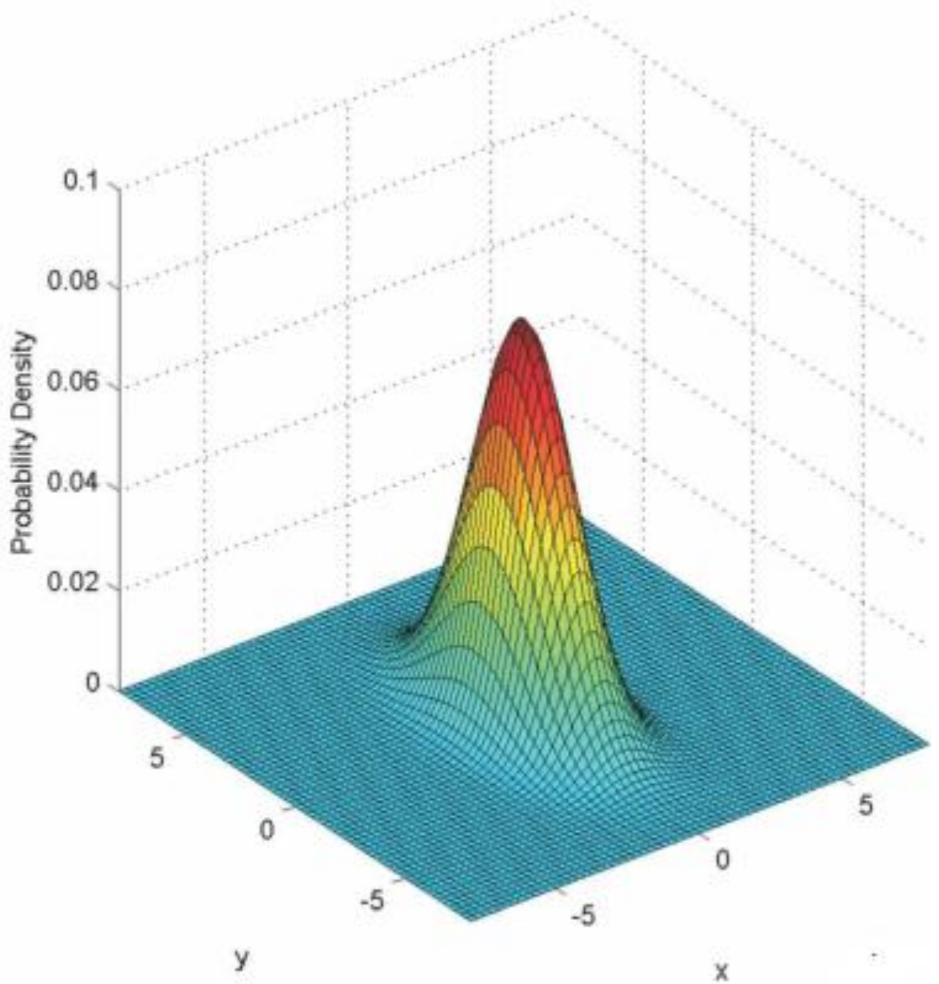
Recap - Independent Normal

$$f_{X,Y}(x,y) = f_X(x)\ f_Y(y)$$

$$= \frac{1}{2\pi\sigma_x\sigma_y}\ e^{\left\{-\frac{(x-\mu_x)^2}{2\sigma^2_x} - \frac{(y-\mu_y)^2}{2\sigma^2_y}\right\}}$$

$\mu_x = \mu_y = 0;\ \ \sigma^2_x = \sigma^2_y = 4;$





The Bayes Rule

<u>Discrete Setting</u>

$X \longrightarrow Y$

unobserved value x

Prior $P_X(.)$

observed value y

Model $P_{Y|X}(.|.)$

Think of a Digital Communication Signal

x

unknown state

noisy observation of y

Inference $P_{X|Y}(.|y)$